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Multiple Moving Objects Tracking for Automated Visual Surveillance

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***Abstract*— Moving objects tracking is of great significance for automated visual surveillance. Conventional tracking algorithms, such as Kalman filter or particle filter, have shown the effectiveness and robustness in many practical applications. However, the Bayesian filter is not designed for tacking multiple moving objects. The difficulty is the data association between the measurements and the tracks. Tracking can fail due to the confusion of similar measurements from adjacent moving objects. This paper proposes an approach for multiple moving objects tracking. We formulate the measurement assignment process as a problem of finding the matching with the maximum weight in a bipartite graph. Moving objects are detected by background subtraction. We test our approach using public datasets. The experimental results demonstrate that our approach is able to track multiple moving objects correctly.**

I. Introduction

Tracking means continuously estimating the positions of a moving object[1]. In other words, tracking aims to obtain the trajectories of the moving object. Tracking of moving objects plays an important role in the applications of the automated visual surveillance. The trajectories of moving objects can be used to analyze the motion behaviors[2]. In the past decades, lots of tracking algorithms have been proposed[3]. The framework of these algorithms, such as Kalman filter or particle filter, are based on the Bayesian filter[4]. The general process of the Bayesian filter is firstly predicting the positions of moving objects using velocities and control information, and then updating the prediction using measurement information. Normally, the control information is given and the velocities can be computed from previous states. However, the measurement information relies on instruments to autonomously estimate the positions of moving objects[5]. For example, in vehicle tracking applications the measurement information can be retrieved from GPS signals. Due to noises or unpredictable reasons, the measurement information can be lost. The tracking algorithms can recover from the lost states through the recursive equations. It is obvious that sufficient and precise measurement information is beneficial for the moving objects tracking.

In visual tracking applications, the positions are the coordinates of moving objects pixels in images[6]. Thus, the motivation of visual tracking is to continuously estimate the coordinates of moving objects. The measurement information for visual tracking can be obtained from motion detection methods. We can roughly classify motion detection methods into two categories[7]. One type of methods require the prior knowledge for the detection initialization. A common type of prior knowledge is the manual or natural features. The features can be corner feature points or a histogram of a patch of pixels[8]. The other set of approaches does not require the prior knowledge. Moving objects can be localized using motion detection algorithms such as optical flow[9] or background subtraction[10] algorithms. The advantage of the first type of approaches is that it can be used for videos captured on moving platforms. Due to the intrinsic limitations of the motion detection algorithms, the second type of approaches can only be used for videos captured by static cameras. However, in automated visual surveillance applications the latter type of approaches is more appropriate[11]. This is because most surveillance cameras are static and the prior knowledge is commonly not available. Thus, in this paper we adopt the latter approach and concentrate our work for surveillance videos captured by static cameras.

The remainder of this paper is organized as follows. In section II, a review for motion detection and tracking is presented. In section III, we discuss the details of our approach. In section IV, the experimental results are presented. In the last section, this paper comes into conclusion and the future work is discussed.

II. Related Work

In this section, we generally overview the motion detection and visual tracking algorithms. Optical flow is a set of algorithms which can detect and track moving objects in video sequences[12]. The approaches of optical flow can be divided into two categories. One category is feature-based, which is called the sparse optical flow. The other is dense optical flow which is base on pixel intensity values. These two types of approaches have been both extensively studied in the past decades.

The sparse methods extract features from consecutive image frames. The feature points can be matched by finding the minimum distances of the feature points in the two image frames[13]. Moving objects can be indicated by the correspondences with the distances larger than a certain threshold. Many robust and effective feature detection and description algorithms[14], such as SIFT, SURF and BRISK[15], have been developed. The computation of the sparse optical flow is fast but it cannot segment the moving objects at the pixel level. Only some pixels of the feature points can be classified. The sparse methods cannot provide precise motion detection information for visual tracking of deformable objects. Different from the sparse methods, the dense methods calculate the optical flow for every pixel in the images. The intensity values of a patch of pixels in consecutive frames

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are employed for the calculation. A typical dense optical flow algorithm proposed by Farneback[16] is based on the polynomial expansion. The algorithm approximates a patch of pixels as a quadratic polynomial expression. The same patch can be found in the consecutive frame. The motion distance of the pixel patch can be found by comparing the coefficients of the two polynomial expressions. Similar as the sparse methods, the moving objects can be indicated by the pixels with larger optical flow values. The dense optical flow algorithms can provide pixel-level segmentation results for moving objects. However, the optical flow values are sensitive to the illumination changes and small movements, so it cannot provide precise motion detection results.

Instead of using optical flow, many visual tracking approaches adopt background subtraction algorithms for moving objects detection. Background subtraction is a kind of motion segmentation approaches which can build the background model and produce the motion masks at the pixel level. The moving objects can be segmented by subtracting the current frame with the background model. There are two types of methods to build the background model. One type is parametric approaches[17], which model each pixel of the background in an analytic expression. The other is non-parametric approaches[18], which model the pixels using a set of samples. The running average-based background subtraction is a typical parametric approach. The pixel value in the background model is a weighted average of the pixel values in the current frame and the background model.

ViBe is an effective non-parametric background subtraction approach[19]. To classify a pixel of the current frame, the algorithm compares the pixel value of the current frame with the corresponding sample values in the model. The pixel can be classified as foreground when the number of distances beyond a predefined radius is greater than a specified threshold. The values of the radius and the threshold are chosen empirically. The algorithm employs a conservative update strategy. It only allows pixels that are identified as background to be used to update the background model. The updating is based on a random policy. Both the pixel and its neighbors are subjected to be updated. The advantage of the ViBe algorithm is the fast model initialization mechanism[20]. Unlike other non-parametric approaches, ViBe initializes the pixel model just using one image frame. This is based on the observation that neighboring pixels share similar intensity values.

Bayesian filter has been widely used in real tracking applications[21]. The recursive equations can predict and update the states of moving objects. There exist lots of variants in different scenarios. The variants can be generally divided into three categories, Single Hypothesis Localization (SHL) Methods, Multiple Hypothesis Localization (MHL) Methods and a hybrid of them. SHL localizes a moving object using one track candidate, such as the Kalman filter. MHL localizes a moving object using multiple track candidates, such as the particle filter[22]. Kalman filter and particle filter are both classical tracking algorithms which have been widely used in many real applications. In some specific scenario, such as tracking objects with abrupt accelerations, a hybrid method is more appropriate. For example, the hybrid tracker of the particle filter and the mean shift algorithm is developed to solve the tracking problem in such case[23]. However, the tracking algorithms are designed for single moving object. An intuitive idea to extend these algorithms for multiple moving objects tracking is using the measurement assignment algorithms. The multiple moving objects tracking problem can be solved by associating the measurements to the most similar predicted positions.

III. Multiple Moving Objects Tracking

A. Overview of Approach

This section presents an overview of our approach. We use kalman filter for objects tracking. The *track* in this paper means a tracking entity which includes the tracking algorithm and the state variable of the moving object. Each moving object is assigned a *track*. As shown in Fig.1, *Movement Prediction* performs the time update equation of the Kalman filter. *Tracks States* are updated by the assigned measurements after the data association. The measurements are the positions of the moving objects. The positions are denoted by the centers of the detected blobs. We use the Pixel-Based Adaptive Segmenter (PBAS)[24] algorithm as the background subtractor. The salt and pepper noises are reduced by the median filter that is recommended in the original paper. In order to get connected pixel blobs, the morphological operations are applied on the background subtraction results.

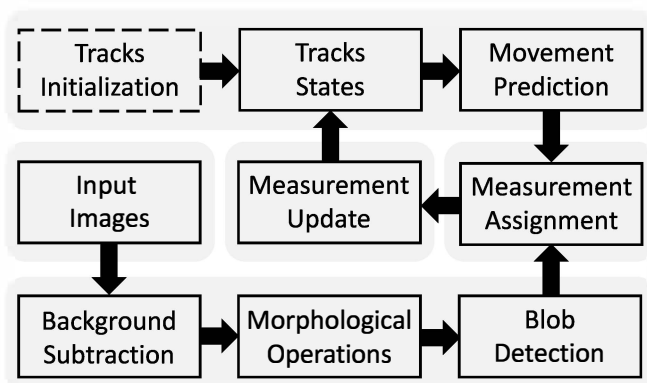


Fig. 1. The schematic overview of our approach. The dashed box indicates the initialization only running in the first loop. *Tracks States* includes the positions and velocities, which also denotes the tracking results. The *Measurement Assignment* problem is solved by the Hungarian algorithm.

B. Kalman Filter

In this paper, the state variable $\boldsymbol{x}$ includes the coordinates and velocities of a moving object. It is expressed as follows,

$$\boldsymbol{x} = [x,\ y,\ v_x,\ v_y]^T, \tag{1}$$

where x and y are the coordinates, v_x and v_y are the velocities. Every moving object is assigned with one state variable. The tracking outputs are the coordinates of the moving objects. The Kalman filter is created with 4 state parameters and 2

measurement parameters. The positions are initialized using the first measurement. The velocities are assigned a random integer value between -1 and 1. With the assumption of discrete-time process, the prior estimated state variable $\hat{\boldsymbol{x}}_k^-$ can be updated using the following equation,

$$\hat{\boldsymbol{x}}_k^- = A\hat{\boldsymbol{x}}_{k-1} + B\boldsymbol{u}_{k-1} + \boldsymbol{w}_{k-1}, \tag{2}$$

where $A \in \mathcal{R}^{4\times4}$ is the transition matrix, $\boldsymbol{w} \in \mathcal{R}^{4\times1}$ is the process noise which is Gaussian distributed, B is the optimal control matrix but we have no control input $\boldsymbol{u}$ in this paper. Equation (2) is also called the time update equation.

Let $\boldsymbol{x}_k$ denote the real value of the state variable, the measurement information $\boldsymbol{z}_k$ can be modelled as follows,

$$\boldsymbol{z}_k = H\boldsymbol{x}_k + \boldsymbol{v}_{k-1}, \tag{3}$$

where $\boldsymbol{v} \in \mathcal{R}^{4\times1}$ is the Gaussian distributed measurement noise. The real value of the state variable can never be known. The Kalman filter aims to estimate the state variable close to the real value as much as possible. With the measurement information, the posterior estimated state variable $\hat{\boldsymbol{x}}_k$ can be calculated as follows,

$$\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}_k^- + K_k(\boldsymbol{z}_k - H\hat{\boldsymbol{x}}_k^-), \tag{4}$$

where $H\hat{\boldsymbol{x}}_k^-$ is the predicted measurement, $K_k \in \mathcal{R}^{4\times2}$ is the Kalman Gain. Equation (4) is also called the measurement update equation.

With the assumption that the state variable is Gaussian distributed, namely $p(\boldsymbol{x}_k) \sim \mathcal{N}(\hat{\boldsymbol{x}}_k, P_k)$, the Kalman Gain K_k can be calculated by the following formula,

$$K_k = \frac{P_k^- H^T}{HP_k^- H^T + R}, \tag{5}$$

where $P_k \in \mathcal{R}^{4\times4}$ is the covariance matrix of the state variable, $R \in \mathcal{R}^{2\times2}$ is the covariance matrix of the measurement noise.

C. Bipartite Graph Matching

Tracking multiple moving objects is challenging because the tracks cannot discriminate the measurements from different moving objects. The key idea of solving this problem is to assign a measurement to the most similar predicted position. The assignment occurs after the time update but before the measurement update. We formulate the assignment process as a problem of finding the matching with the maximum weight in a bipartite graph[25]. It can be solved by the Hungarian algorithm[26]. Let's consider a bipartite graph G,

$$G = (U, V, E), \quad U \cap V = \emptyset, \tag{6}$$

where $U = \{\hat{\boldsymbol{x}}_1^-, ..., \hat{\boldsymbol{x}}_i^-, ..., \hat{\boldsymbol{x}}_N^-\}$, $V = \{\boldsymbol{z}_1, ..., \boldsymbol{z}_j, ..., \boldsymbol{z}_M\}$, N and M are the numbers of vertices in the two sets. The edges in E connects the vertices in U and V. A matching $M \subset E$ is a subset in which no two edges share a same vertex. The problem is to find a set W which has the maximum weight. The edge weight measures the similarity between two vertexes. Fig.2 shows our bipartite graph, the predicted states and the measurements form the two vertex sets U and V. In this paper, the Euclidean distance $d = \left\|T\hat{\boldsymbol{x}}_i^- - \boldsymbol{z}_j\right\|$ between the predicted position and the measured position is employed to evaluate the similarity. The matrix $T \in \mathcal{R}^{2\times4}$ is used to extract the coordinates from the state variable $\hat{\boldsymbol{x}}_i^-$. The Gaussian kernel is used to calculate the edge weight c_{ij},

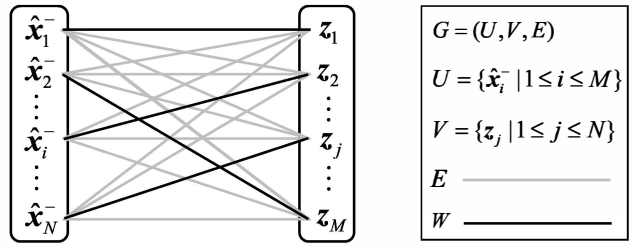


Fig. 2. The bipartite graph for the measurement assignment problem. The gray edges denote the set E. The black edges denote the sample matching set W which has the maximum weight.

$$c_{ij} = \alpha \exp\left[\frac{(T\hat{\boldsymbol{x}}_i^- - \boldsymbol{z}_j)^2}{2\sigma}\right], \tag{7}$$

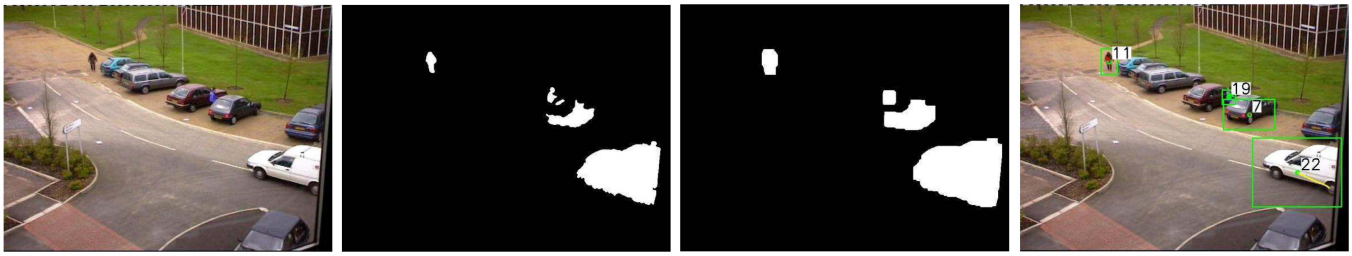
where σ is the standard deviation of the Gaussian kernel. The constant α can make sure all the weights sum to 1.

D. Tracks Management Mechanism

Tracks are initialized using the first measurement. The number of measurements determines the number of the initialized tracks. The initial positions of the tracks are the first measured moving objects positions. The tracking may be lost in some cases, such as moving objects disappearance or occlusions. We introduce a variable *lostCount* for each track to count the number of lost frames. *lostCount* is increased in each loop when the track cannot be assigned a measurement or the distance between the predicted position and the assigned measurement is greater than a threshold. The track will be deleted when the variable *lostCount* is greater than a threshold. In the case that there exist some measurements that are not assigned to any track, new tracks will be created because this usually happens when new moving objects appear in the scene.

IV. Experimental Results and Discussion

We implement our approach on a PC with Intel i3 CPU and 4GB memory. The experiments are performed using real video surveillance datasets *PETS2000* and *PETS2003*. In the videos, the camera remains static and no dynamic background exists. In *PETS2000*, a car is moving and some people are walking on a street. In *PETS2003*, the video records a football game. We are aiming to track the car, the walking people and the football players in the videos. The moving objects are detected by the PBAS background subtraction algorithm. The masks produced by PBAS are denoised by the median filter. The erosion and dilation are applied sequentially on the masks. The structuring elements for the erosion and dilation are 10×10 and 25×25 squares respectively. We use the blob detector in *OpenCV* to get the coordinates of the blob center.



(a) (b) (c) (d)

Fig. 3. The demonstration of our tracking process using a frame in *PETS2000*. (a) is the original image frame. (b) is the result of the PBAS background subtraction algorithm. The mask has been denoised using the median filter. (c) is the morphological operation result. The blob detection algorithm is applied on (c) to get the moving object positions, which are used as the measurement information for the Kalman filter. (d) is the tracking result. The green circle and the green rectangle denote the estimated center and the corresponding bounding rectangle of the moving object. The text label shows the tracking ID. The traces with random colors record the estimated center positions in the recent frames.

Fig.3 shows the moving objects tracking process. The PBAS algorithm produces the moving objects segmentation results, which are used as the motion detection information. The masks are denoised using the median filter which is recommended in the original paper. From figure (b), we can see the PBAS results cannot exactly indicate the number of moving objects. Thus, the erosion operation is applied on the PBAS masks to remove small fragments of the moving objects. The dilation is applied after the erosion operation to make sure there are pixel blobs that can represent the moving objects. The detected blob center positions provide the measurement information for the Kalman filter. Figures (a)-(c) show the process of producing the measurement information. The measurements are assigned for different tracks using the bipartite graph matching. With the assigned measurements and predicted positions, Kalman filter produce the estimated positions of the moving objects, which are denoted as green circles in figure (d). From the traces of the moving objects, we can see No.7 has stopped for a while. It is easy to find that No.19 is a new moving object by observing the adjacent frames. The green rectangles in figure (d) are the bounding rectangles of the detected blobs, which can roughly denote the moving objects.

Fig.4-Fig.5 show the sample tracking results of the testing video sequences of *PETS2000* and *PETS2003*. The figures demonstrate that our approach is able to track multiple moving objects correctly. In Fig.5, the No.28 football player is appearing gradually. We can see our approach is able to create new track for the new moving object. Because the red shirt football player moves quite slowly, the erosion operation erases the corresponding small patch of pixels in the mask produced by the PBAS algorithm.

V. Conclusions

This paper proposed an approach of multiple moving objects tracking for automated visual surveillance. The moving objects are detected and segmented by the PBAS background subtraction algorithm. The key idea for tracking multiple objects is formulating the measurement assignment process as the problem of finding the matching with the maximum weight in a bipartite graph. The Kalman filter is employed for the information fusion of the predicted positions and the assigned measurements. Due to the use of background subtraction, our approach is limited for the videos captured by static cameras. Thus, in the future work, we would like to extend our approach for surveillance videos captured by moving cameras.

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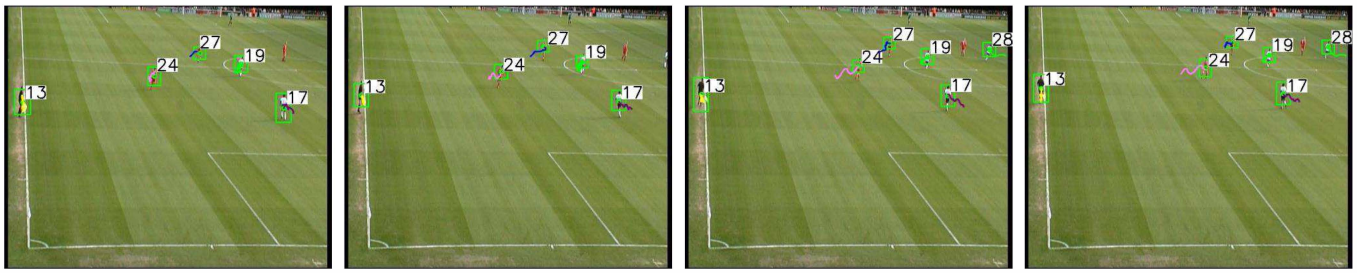
(a) (b) (c) (d)

Fig. 4. The sample tracking results for the *PETS2000* testing dataset. (a)-(d) are the frames No.442, No.500, No.551 and No.620 respectively. The walking people and the moving car denoted by the green rectangles are tracked in the image sequences.



(a) (b) (c) (d)

Fig. 5. The sample tracking results for the *PETS2003* (Testing-Camera4) dataset. (a)-(d) are the frames No.341, No.350, No.372 and No.380 respectively. The football players are tracked in the image sequences.

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